

CIVL4340

Tutorial Exercises – Week 1 & 2

The objectives of this tutorial are:

1. To download a large (>250 MB) dataset of wind observations and determine how you can open and interact with these data.
2. Undertake some preliminary analysis on your dataset.

The Bureau of Meteorology measures weather data at Automatic Weather Stations (AWS) located around Australia and its territories (including Antarctica). AWS are a primary tool used by engineers and meteorologists to understand long term weather and climate around the country. Wind engineers use wind data collected at AWS to define the wind climate at a specific location to aid in defining appropriate wind speeds to use for ultimate limit and serviceability limit state design. Historic AWS records can be sourced from the Bureau of Meteorology, with recent (last 72 hours) observations available to view on their website (e.g. recent Queensland data can be viewed [here](#)).

Each student has been assigned an AWS location to analyse (see table at end of this document). Once you have identified your station, download the data file from the OneDrive folder [here](#). The last six digits in the file name indicates the BOM ID. Please make sure you download the correct data file as this is the one you will end up using for Assignment 1 and these exercises will help you better understand your data.

Once you have downloaded your data, do the following and answer any questions posed.

1. Open your data file using a program that will allow you to analyse and manipulate the data contained within it. Note that these files are large (>250 MB), so Excel may not be your best option. Data in your file is comma separated and contained within 5 columns labelled, *tm*, *WSMean*, *WSMax*, *WDir*, *WDirSD*, where *tm* is the date/time of record, *WSMean* is the mean wind speed measured over the previous 1-minute period, *WSMax* is the maximum 3-second gust within the previous 1-minute period, *WDir* is the mean wind direction, and *WDirSD* is the standard deviation of wind direction. All wind speeds are provided in km/h.
2. Identify the ten highest wind gusts within your data set, extracting all available variables for these gusts. Look at the time of these gusts and wind direction occurring at that time. Can you draw any conclusions about what season causes the strongest gusts at your station and what direction they may come from? Explain your findings in a few sentences.
3. Create a data table that lists the maximum wind gust that occurred during each month of your record. Create a plot that shows these data and write a short few sentences explaining how the gust wind climate varies for each month of the year.
4. If we consider that wind gusts greater than 20 m/s have the potential to cause damage to buildings or interruption to infrastructure operation, determine how often this damage threshold is exceeded each year of your record and create a plot to

show your results. Are you able to see any temporal trends in your data? Indicate any trend on your plot and write a few sentences explaining what you have found.

5. If we want to understand the direction strong wind gusts (use 20 m/s again as a threshold) typically come from, determine how to manipulate your data to answer this question. Are there any dependencies on the season in these observations? Again, create a plot/s to show your answers and write a few sentences explaining your analysis.

*Please ensure you practice making figures that are easy to read and interpret. This includes ensuring you appropriately label all axes and include units where appropriate.

Table 1. Weather station allocation

BOM ID	Station Name	Latitude	Longitude	State	Year start	Year end	Student ID 1	Student ID 2
31011	CAIRNS AERO	-16.8736	145.7458	QLD	2001	2024	48923464	46985516
32040	TOWNSVILLE AERO	-19.2483	146.7661	QLD	2001	2024	48829379	49089415
33119	MACKAY M.O	-21.1172	149.2169	QLD	2001	2024	48018995	48033112
36031	LONGREACH AERO	-23.4397	144.2828	QLD	2001	2024	47484780	48091653
29039	MORNINGTON ISLAND	-16.664	139.1837	QLD	2003	2015	48014876	48012603
29127	MOUNT ISA AERO	-20.6778	139.4875	QLD	2003	2024	47440731	48509596
31192	GREEN ISLAND	-16.7614	145.9719	QLD	2003	2010	46962843	47488461
27045	WEIPA AERO	-12.6778	141.9208	QLD	2004	2024	50087435	47451090
33045	MACKAY AERO	-21.1706	149.1794	QLD	2004	2024	48037802	48038283
33317	HAY POINT	-21.2894	149.3008	QLD	2005	2024	50115552	49816433
27058	HORN ISLAND	-10.5844	142.29	QLD	2009	2024	48040905	48026066
32182	CAPE FERGUSON NTC AWS	-19.2774	147.0586	QLD	2009	2024	47436145	48031677
33208	ROSSLYN BAY NTC AWS	-23.161	150.7901	QLD	2009	2024	46978394	48026132
36034	BLACKALL AIRPORT	-24.4303	145.4306	QLD	2009	2024	48102942	48025331
27075	SCHERGER RAAF	-12.6167	142.0869	QLD	2010	2024	46986157	47859173
31037	LOW ISLES LIGHTHOUSE	-16.3842	145.5592	QLD	2010	2024	47439100	48104731
33308	SAMUEL HILL AERO	-22.7433	150.6578	QLD	2010	2024	49813973	47021354
35124	CLERMONT AIRPORT	-22.7756	147.6217	QLD	2010	2024	48036225	47472741
35129	ROLLESTON AIRPORT	-24.4617	148.6264	QLD	2010	2024	47847899	46976934
35139	LOCHINGTON	-23.9424	147.525	QLD	2010	2024	46416485	47424982
27054	COCONUT ISLAND	-10.0511	143.0686	QLD	2011	2024	47425260	47496028
27073	COEN AIRPORT	-13.7606	143.1183	QLD	2011	2024	49093506	48029209
28008	LOCKHART RIVER AIRPORT	-12.785	143.3047	QLD	2011	2024	48022424	47981935
29038	KOWANYAMA AIRPORT	-15.4818	141.7483	QLD	2011	2024	46414865	49321049
29058	JULIA CREEK AIRPORT	-20.6672	141.7214	QLD	2011	2024	48027214	47650286
29077	BURKETOWN AIRPORT	-17.7483	139.5356	QLD	2011	2024	48010252	47428636
29141	CLONCURRY AIRPORT	-20.6664	140.505	QLD	2011	2024	48828608	48025836
29167	CENTURY MINE	-18.7569	138.7056	QLD	2011	2024	47246504	47451399
30022	HUGHENDEN AIRPORT	-20.8192	144.2333	QLD	2011	2024	48022330	48817198
30124	GEORGETOWN AIRPORT	-18.3039	143.5306	QLD	2011	2024	48019769	48774022
30161	RICHMOND AIRPORT	-20.7001	143.1137	QLD	2011	2024	47453160	48040446
31209	COOKTOWN AIRPORT	-15.4461	145.1861	QLD	2011	2024	48009340	47432615
31210	MAREEBA AIRPORT	-17.0704	145.4293	QLD	2011	2024	49052772	46975450
31213	CAPE FLATTERY	-14.9672	145.3106	QLD	2011	2024	49061877	48013440
32037	SOUTH JOHNSTONE EXP STN	-17.6053	145.9972	QLD	2011	2024	46280444	48092632
33002	AYR DPI RESEARCH STN	-19.6169	147.3758	QLD	2011	2024	48033336	47383052
33106	HAMILTON ISLAND AIRPORT	-20.3658	148.9536	QLD	2011	2024	47427376	46823261